

Mubashir Rehman

Backend Engineer · Python / Django / FastAPI · REST APIs & Automation · PostgreSQL / Redis / Docker

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SUMMARY

Backend-focused Software Engineer with **3+ years** building production Python systems: REST APIs, PostgreSQL schemas, background workers and their deployment, across healthcare integration, HRMS/ERP, industrial estimation and AI/LLM automation. Co-author of a peer-reviewed Springer paper (CSSP, 2025).

PUBLICATION

Evaluation of a low-cost single-lead ECG module for vascular ageing prediction and studying smoking-induced changes in ECG

Circuits, Systems, and Signal Processing (Springer), 44(8), 5852-5875, 2025 · DOI: 10.1007/s00034-025-03048-2

Co-author, equal contribution (3rd of 7 authors): data collection and the ML pipeline. **42 subjects, 6,131 segmented samples**; random forest **R² 0.99** on segmented data and **0.87** with transfer learning on unsegmented data.

EXPERIENCE

Software Engineer, TransData · Lahore, Pakistan · Mar 2025 to Present

- Became top contributor (**272 of 389 commits**) on a React 19 and Node.js/TypeScript operations console for an HL7 healthcare integration engine, by building its BFF API layer: bulk message reprocess and export with CSRF-protected multipart handling, PHI-access auditing features, and real-time message-trend dashboards.
- Delivered, as sole engineer, a customised Django HRMS covering **138 employee records and ZKTeco biometric sync across 2 devices**, by designing a multi-device attendance schema, day- and month-level reporting endpoints, payroll calculation logic, Celery jobs, and the production CI/CD pipeline.
- Let prompt changes ship without a code deploy (**28 prompts across 11 workflows** migrated), as top contributor (191 of 268 commits) on a LinkedIn content-scheduling platform, by replacing an external workflow tool with an in-house, versioned, JSON-backed prompt system and a hierarchical admin UI.
- Took an industrial parts cost-estimation platform from unstable to stable and **handed it over to another team member**, by owning its backend (normalised PostgreSQL schemas, authenticated REST APIs serving the AI/ML quoting service, Celery ingestion) and its CI/CD and Docker environments.
- Built a crypto and social-intelligence backend end to end as sole author (**154 of 155 commits**) and **handed it over to the client**, by writing Telegram, Discord and Twitter/X ingestion, AI summarisation, pgvector semantic search on Gemini embeddings, and an alert engine with ARQ background tasks on Redis.

Software Engineering Team Lead, VeritusLabs · Oct 2023 to Mar 2025 (Software Engineer Jul 2023 to Oct 2023)

- Promoted to Team Lead and led **a team of 4 engineers** on QGroundControl-based drone-control and swarm features in C++ and Qt/QML, including mission planning and real-time telemetry over MQTT.
- Delivered a full-stack AI prompt-management web app with streaming OpenAI responses, RBAC and S3-backed storage, **deployed through Docker and CI/CD**, by building its Django/FastAPI backend with LangChain; also replaced manual AWS console setup with Boto3 scripts provisioning VPC, EC2, IAM and ECS.

PROJECTS

HireTrack · React 19, TypeScript, LangGraph, Supabase · github.com/mubashir-rehman/job-application-tracker

- Built and open-sourced a job-application tracker (sole author, **89 commits, public repo, live deployment**), by shipping a deterministic-first LangGraph pipeline that tailors resumes and scores job descriptions across five AI providers (bring-your-own-key), plus an offline-first PWA and a remote MCP server.

SKILLS

Languages: Python, SQL, TypeScript, JavaScript, C++, Bash

Backend and data: Django, Django REST Framework, FastAPI, Flask, Node.js, REST API design, OAuth2, JWT, RBAC, PostgreSQL, MySQL, Redis, pgvector, SQLAlchemy, Alembic, Celery

AI/LLM: OpenAI API, Gemini, LangChain, LangGraph, embeddings, semantic search, vLLM, LiteLLM

DevOps: Docker, Nginx, Traefik, GitHub Actions, CI/CD, AWS (EC2, ECS, VPC, IAM, S3), DigitalOcean, Linux

Enterprise: ERPNext/Frappe, Horilla HRMS, ZKTeco biometric API, HL7 integration

EDUCATION

BS Computer Science, Information Technology University, Lahore